

WATERMILLS — HOW THEY WORK

Water flows onto the wheel's blades or buckets, the wheel turns (typically **10-20 rpm**), and a gear train multiplies the speed up to millstone speed (**100-150 rpm**). The classic drive chain:

Water wheel on the axle turns the **pit wheel** (large vertical gear).
Pit wheel drives the **wallower** - a small lantern gear on a horizontal shaft (first speed step-up, ~2-4x).
The wallower's shaft drives the **great spur wheel** (horizontal, like a crown gear).
A **stone nut** (pinion) on each millstone meshes with the spur wheel - the stone spins at 100-150 rpm.
Grain drops from the **hopper** through the shoe into the **eye** (centre hole) of the rotating **runner stone** .
The runner grinds grain against the fixed **bed stone** ; flour is flung out at the rim and falls down the meal spout.

Wheel types (efficiency = how well water energy is captured)

Type	How water hits the wheel
Undershot	Flat blades dip into the stream below - simplest, works in low-head/high-flow rivers
Breastshot	Water meets the wheel at axle height (mid-level)
Pitchback	Water enters at the top but turns the wheel backwards (self-cleaning buckets)
Overshot	Buckets fill at the top from a flume/launder - gravity does the work, needs ~2 m head
Poncelet	Curved blades, undershot position, redirects flow for extra push

Power of a waterwheel: $P = \text{density} \times g \times \text{flow} \times \text{head} \times \text{efficiency}$, i.e. **$P \text{ (watts)} = 1000 \times 9.81 \times Q \text{ (m}^3\text{/s)} \times h \text{ (m)} \times \text{efficiency}$** . Worked example: $Q = 0.5 \text{ m}^3\text{/s}$, $h = 3 \text{ m}$, 75% efficient -> $P = 1000 \times 9.81 \times 0.5 \times 3 \times 0.75 \sim 11,000 \text{ W} = \sim 15 \text{ hp}$ - enough for 4 pairs of stones (one pair needs 3-10 hp).

Beyond flour: fulling cloth (trip hammers), sawing timber, crushing ore (stamp mills), pumping water (Archimedes screw), blowing furnace bellows, and powering Arkwright's cotton mills (Cromford, 1771).

MODERN MILLS — SAME CHAIN, NEW POWER

The capture -> rotate -> gear -> tool chain survives in every modern industry:

Flour - roller mills (1870s): fluted steel **break rolls** crack the grain, sifters separate bran/germ/endosperm, smooth **reduction rolls** grind to flour; continuous, hygienic, ~75% extraction - >95% of world flour today.
Paper : the **Fourdrinier machine** (1807) - pulp slurry onto a moving wire mesh, water drains, press rollers squeeze, heated drying cylinders finish - paper rolls at 1,500+ m/min.
Steel : **rolling mills** squeeze red-hot ingots through successive roller stands (hot rolling), then cold rolling for precision - sheets, rails, beams for the modern world.
Sawmills : band saws & circular saws, laser scanning, debarking, chipping - computerised from log to plank.
Textiles today : automated ring spinning and air-jet looms (1,000+ picks/min) - a mule shed that needed 500 workers now runs with 10.
Hydroelectric : dammed 'head' drives Pelton (high-head), Francis (medium), Kaplan (low-head) turbines - ~16% of world electricity.
Wind turbines : 3-blade HAWT, 100+ m rotor, gearbox -> generator, still Betz-limited like their ancestors - ~7% of world electricity and climbing.

pit wheel driving an 8-stave wallower steps speed up 6x; a spur wheel or 60 teeth driving a 15-tooth stone nut steps up 4x more. Chain: 12 rpm wheel x 6 x 2.5 ~ 180 rpm at the stone. Gear teeth were cut from wood and iron long before precision metal-cutting existed - millwrights were the first mechanical engineers.

turning; **Roller reefing** (1950s): fabric rolled up like a blind.

Power of a windmill: $P = 0.5 \times \text{air density} \times \text{swept area} \times \text{wind speed}^3 \times C_p$. Air density ~1.225 kg/m3, area = $\pi \times r^2$, and C_p (rotor coefficient) is at most the **Betz limit 0.593** - no rotor can capture more than 59.3% of wind energy. Worked example: $r = 10 \text{ m}$, $v = 8 \text{ m/s}$, $C_p = 0.4$ -> $P = 0.5 \times 1.225 \times 314 \times 512 \times 0.4 \sim 39,400 \text{ W} \sim 53 \text{ hp}$. Note the **v^3** : doubling wind speed gives 8x the power - location is everything.

Famous jobs: draining Dutch **polders** (land reclaimed below sea level, pumped by wind-driven Archimedes screws - ~17% of the Netherlands was won from the sea), grinding grain, sawing timber, pressing oil and paper-making.

FUN FACTS

The **Domesday Book** (1086) counted 5,624 watermills in England - more mills per person than most of Europe.
Barbegal (~300 CE): Rome's 16-wheel industrial estate ground ~25 tonnes of flour a day for Arles.
The **first windmills** (Persia, 7th c.) spun on a **vertical axis** - a completely different design from today's.
Mill idioms everywhere: **'run of the mill'**, **'grist to the mill'**, **'millstone around your neck'** .
The **mill governor** was the first feedback device (ancestor of the thermostat) - Watt copied it for steam engines, and 'milling' a coin's edge stopped silver-clipping.
Windmills drained the sea and created ~ **17% of the Netherlands'** current land area.

brake wheel	used for emission, also the brake drum	HAWT	horizontal-axis
Cap	Rotating roof of tower/smock mills	Pelton / Francis / Kaplan	High / medium / low-head

power)

heavy, later, then heads and stepped wheels as young people - the end of steam law.

Impact: cotton exports multiplied ~100x in a century; canals, railways and mill towns followed; capital investment and wage labour replaced cottage work. The mill gave the world the **modern factory, shift work and mass production** .